

ML Use Cases & Python Setup

Student Assessment — 51 questions. Answer everything, then click **Submit** at the bottom to see model answers.

Topic 1: ML Use Cases & Problem Framing

Section A: Identify the ML Problem Type

Q1

An e-commerce company wants to predict whether a customer will return a purchased product.

- Classification / Regression / Clustering?
- What is the target?
- What features would you use?

Classification

Target =yes/no for customer will return a purchased product

Logistic Regression

✓ Model Answer

Classification (binary). Target: Returned (Yes/No). Features: product category, price, discount, customer past return rate, days-to-return-window, review/rating left, payment method, shipping time, size/color variant, repeat customer flag.

Q2

A bank wants to estimate how much loan amount a customer is likely to repay.

Bank Should Check Customer Salary that it is in his capabilities and also check cibil score for approval and to trust the customer.

✓ Model Answer

Regression task. Target variable: repaid amount (continuous, e.g. in € /\\$). Features: income, credit score, employment status, existing debts, loan tenure, past repayment history, age, collateral value.

Q3

A streaming platform wants to recommend movies to users.

- Classification? Recommendation? What data?

Classification is used for this movie recommendation system mostly and it depend on users interest also which movies they are watching and recommendation system will be suggest same genre movie or new genre movie for customer which might they like. they have

✓ **Model Answer**

It is a Recommendation problem (not a standard classification, though some recommenders cast it as classification of like/no-like). Data: user watch history, ratings, watch duration, genre/actor metadata, time-of-day, similar users' behavior, search history.

Q4

A manufacturing company wants to predict machine failures before they occur.

there will be use of forecasting or future predict based on data available. for manufacturing company there is product like toy truck it will stop working after short time because of its overuse and heat motor or any malfuntion or remote not working that is defect that knew it

✓ **Model Answer**

Business problem: reduce downtime & maintenance cost (predictive maintenance). ML problem: binary classification (will fail in next N hours: Yes/No) or time-to-failure regression / survival analysis. Features: sensor readings (vibration, temperature, pressure), machine age, last maintenance date, runtime hours, load.

Q5

A retail company wants to estimate next month's sales.

Well, Sales is not something you can predict but we do forecasting for this product and we know data based on past data and market strategy for product how they create hypo of it for customer to use it. and they know customer will buy product and they what they are

✓ **Model Answer**

Time-series forecasting / regression. Useful metrics: MAE, RMSE, MAPE. Features: historical sales, seasonality, promotions, holidays, inventory, weather, competitor pricing.

Section B: Convert Business Problem → ML Problem

Q6

A telecom company is losing customers every month. Frame as Business Objective, ML Task, Target, Features, KPI.

Telecom Company should analyze why there are losing customer every month by
1) business objective - reduce customer losing
2) ml Task - analyze why there are leaving and give report.

✓ **Model Answer**

Objective: reduce monthly churn rate. ML Task: binary classification (churn in next 30 days). Target: Churned (Yes/No). Features: tenure, monthly bill, plan type, data usage, call-drops, complaints, payment delays, competitor offers. KPI: % churn reduction, retention rate, ROI of retention campaigns. Model metric: Recall / F1 / AUC.

Q7

A hospital wants to identify patients at high risk of being readmitted within 30 days.

patients at high risk should given advice to revisit hospital after 7 seven of discharge for week report how there health is. and to reduce readmit in hospital for them but who have high risk they will admitted to hospital after 30 day because having 2 or 3 stage diagnosis is

✓ Model Answer

Objective: reduce 30-day readmissions. ML Task: binary classification. Target: Readmitted within 30 days (Y/N). Features: age, diagnosis, comorbidities, length of stay, prior admissions, medications, lab results, discharge destination. Metric: Recall + AUC (false negatives are costly).

Q8

An insurance company wants to detect fraudulent claims.

insurance company should do field work by themselves by visiting everything by there executives and process that there no fraudulent claims on company.
if there are new customer and they are asking money to company after within 1 week of

✓ Model Answer

Objective: minimize fraudulent payouts. Target: Fraud (Y/N). Features: claim amount, claim type, time since policy purchase, claim frequency, claimant history, location, doctor/garage involved, mismatched documents. Metric: Precision-Recall AUC, Recall on fraud class (data is imbalanced — accuracy is misleading).

Q9

A ride-sharing company wants to predict trip demand for the next hour.

ride-sharing company use data whether which trip is for based on which season or now having most visited place right now in world. to predict they analyze last years whether at what time what people what. and they release a poster or online post for going in this trip

✓ Model Answer

Regression / time-series forecasting per zone $\times$ hour. Target: number of trip requests. Features: historical demand, time of day, day of week, weather, events, holidays, location/zone, recent surge data. Metric: MAE / RMSE.

Q10

An online education platform wants to identify students likely to drop out.

online education platform predict that how many student are not doing lecture or material study or not attending any live classes of teacher. so they are losing interest in online study they might drop out.

✓ **Model Answer**

Binary classification. Target: Dropout (Y/N) within X days. Features: login frequency, video completion %, assignment submissions, quiz scores, time spent, forum activity, course difficulty, payment status. Metric: Recall / F1.

Section C: Feature Engineering Thinking

Q11

For customer churn prediction, list 10 possible features.

1. login behavior
2. complaints
3. subscription age

✓ **Model Answer**

Tenure, monthly charges, total charges, contract type, payment method, internet/service type, number of support tickets, last complaint date, data usage, plan changes in last 6 months, late payments count, competitor offers received.

Q12

For house price prediction, list 10 possible features.

1. sqfeet area
2. location
3. number of bedrooms

✓ **Model Answer**

Area (sq ft), bedrooms, bathrooms, location/locality, age of property, floor number, furnishing status, parking, amenities, distance to schools/metro, crime rate, recent sale prices nearby.

Q13

For employee attrition prediction, list 10 possible features.

1. monthly income
2. overtime status
3. distance from home

✓ Model Answer

Tenure, salary, last hike %, performance rating, promotions count, manager changes, commute distance, overtime hours, job satisfaction survey, training hours, department, age, education.

Q14

For fraud detection, what features would be useful?

1. transaction amount
2. location
3. device id

✓ Model Answer

Transaction amount, time of day, location, merchant category, device ID, IP address, transaction velocity (txns/hour), distance from last txn, account age, average txn amount, mismatched billing/shipping, unusual hour flag.

Q15

Dataset: CustomerID, Age, Gender, Salary, SubscriptionType, MonthlyUsage. Which are features vs targets?

Features : CustomerId Age, Gender, Salary, Subscription Type,
target : Monthly Usage

✓ Model Answer

CustomerID = identifier (not used as feature). Features: Age, Gender, Salary, MonthlyUsage. Possible targets depending on problem: SubscriptionType (classification — which plan they'll pick) or MonthlyUsage (regression — predict usage). Salary could

also be a target if predicting income.

Section D: Evaluation Metrics

Q16

Which metric for: (a) Spam detection (b) House price (c) Churn (d) Recommendation?

- (a) Spam detection - Precision
- (b) House price - RMSE
- (c) Churn - F1-Score

✓ Model Answer

(a) Precision/Recall/F1 — false positives annoy users. (b) RMSE/MAE/ R^2 . (c) Recall/F1/AUC — imbalanced data. (d) Precision@K, Recall@K, NDCG, MAP, Hit Rate.

Q17

Why can Accuracy be misleading in Fraud Detection?

Accuracy is misleading in fraud detection due to extreme class imbalance.

✓ Model Answer

Fraud datasets are highly imbalanced (e.g. 99% non-fraud). A model predicting 'no fraud' for everyone gets 99% accuracy but catches zero fraud — useless. Use Precision, Recall, F1, PR-AUC instead.

Q18

Fraud dataset: 99% Non-Fraud, 1% Fraud. Model predicts all as Non-Fraud. Accuracy & usefulness?

- ☐ Accuracy 99%, model is excellent
- ☒ Accuracy 99%, but useless — recall on fraud = 0

☐ Accuracy 1%, model is useless

☐ Accuracy 50%, model is mediocre

✓ **Model Answer**

Accuracy = 99% but the model catches zero fraud (Recall = 0). It's useless for the business goal.

Q19

Difference between Precision, Recall, F1. When is Recall more important?

Recall is more important when the cost of missing is positive case(catastrophic).

✓ **Model Answer**

Precision = $TP / (TP + FP)$ — of predicted positives, how many are correct. Recall = $TP / (TP + FN)$ — of actual positives, how many we caught. F1 = harmonic mean of the two. Recall matters more when missing a positive is costly: cancer detection, fraud, predictive maintenance.

Q20

Which metric for medical diagnosis / fraud / recommendation, and why?

1. Medical Diagnosis - recall
2. Fraud Detection - precision
3. Recommendation Systems -mae

✓ **Model Answer**

Medical diagnosis: Recall (missing a sick patient is dangerous). Fraud: Precision-Recall AUC / Recall (catch fraud, but don't drown investigators in false alarms). Recommendation: Precision@K, NDCG (top-K relevance matters more than overall accuracy).

Section E: Industry Case Studies

Q21

Design ML for: predict which leads are most likely to convert.

1. Binary Classification
2. Lead Profile
3. Logistic Regression

✓ Model Answer

Binary classification + probability ranking. Target: Converted (Y/N). Features: lead source, demographics, website behavior, email opens, time-to-first-response, past purchases, company size. Output: probability score; sales team works top-N. Metric: AUC, Precision@K.

Q22

Design ML for: predict employee resignation.

1. Key Features - Financial
2. Algorithm - XgBoost
3. Metrics - recall

✓ Model Answer

Binary classification. Target: Resigned in next 6 months. Features: tenure, salary growth, performance, manager rating, leaves taken, overtime, internal transfers, engagement survey. Output: risk score for HR. Metric: Recall + Precision balance.

Q23

Design ML for: recommend products on e-commerce.

- Core Features - User Data
2. Algorithm - XgBoost
3. Metrics - recall

✓ Model Answer

Recommendation system — collaborative filtering, content-based, or hybrid (often a two-tower / matrix-factorization model + ranking model). Data: purchase history, clicks, cart adds, ratings, product metadata, similar users. Metric: CTR, Precision@K, NDCG, conversion lift in A/B test.

Q24

Design ML for: detect defective products from images.

Core Features - User Data
2. Algorithm - XgBoost
3. Metrics - recall

✓ Model Answer

Computer vision — binary or multi-class image classification (CNN: ResNet, EfficientNet) or object detection if locating the defect. Data: labeled images of good vs defective. Metric: Recall (don't miss defects), Precision (don't reject good units). Deployment: on-edge camera at the assembly line.

Q25

Design ML for: forecast inventory requirements.

Core Features - User Data
2. Algorithm - XgBoost
3. Metrics - accuracy

✓ Model Answer

Time-series forecasting per SKU × location. Models: ARIMA, Prophet, XGBoost, LSTM. Features: historical sales, lead time, seasonality, promotions, holidays, supplier delays. Metric: MAPE / RMSE. Output feeds reorder system.

Topic 2: Python Setup for ML

Section F: Environment & Tools

Q26

Why create a virtual environment for every ML project?

because it needed too safe to run a localhost on any project

✓ **Model Answer**

Isolates dependencies per project — different projects need different library versions (e.g. TF 2.10 vs 2.15). Prevents version conflicts, makes the project reproducible, easier to share via requirements.txt / environment.yml.

Q27

Difference between Python, Anaconda, Miniconda.

Python is the actual programming language, while Anaconda and Miniconda are software distributions that bundle Python with tools to manage code environments and packages

✓ **Model Answer**

Python = the language/interpreter. Anaconda = full distribution bundling Python + 250+ data-science libraries + conda package manager (~3 GB). Miniconda = minimal: Python + conda only; you install what you need (~400 MB).

Q28

What problem does Conda solve?

Conda solves the problem of "dependency hell" and environment fragmentation by acting as both a language-agnostic package manager and an environment manager

✓ **Model Answer**

Manages packages AND environments AND non-Python dependencies (C libraries, CUDA, etc). Pip only handles Python packages; Conda handles the full stack and resolves binary compatibility — important for ML libs like TensorFlow, PyTorch.

Q29

Explain: `conda create -n ml python=3.11`

Type your answer here...

✓ Model Answer

`conda` → the package manager. `create` → make a new environment. `-n ml` → name the environment 'ml'. `python=3.11` → install Python 3.11 in it. Result: an isolated env called 'ml' with Python 3.11.

Q30

What happens when you run: `conda activate ml`

the Conda package manager switches your current shell session to an isolated environment named "ml".

✓ Model Answer

Switches the current shell to use the 'ml' environment — its Python interpreter and installed packages take priority over the base/system ones. Prompt usually shows (ml) prefix. Any pip/conda install now goes into this env.

Section G: Libraries

Q31

Purpose of `numpy`, `pandas`, `matplotlib`, `scikit-learn`, `xgboost`.

The core purpose of these five Python libraries is to handle the complete data science and machine learning workflow

✓ Model Answer

`numpy` → numerical arrays & vectorized math. `pandas` → tabular data manipulation

(DataFrames). matplotlib → plotting / visualization. scikit-learn → classical ML (regression, classification, clustering, preprocessing, metrics). xgboost → gradient-boosted decision trees, top-tier for tabular data.

Q32

Match: (a) Data manipulation (b) Model training (c) Visualization (d) Numerical computing — pick the best single library.

☒ pandas / scikit-learn / matplotlib / numpy

☐ numpy / pandas / xgboost / matplotlib

☐ matplotlib / numpy / scikit-learn / pandas

☐ scikit-learn / pandas / numpy / xgboost

✓ Model Answer

(a) pandas (b) scikit-learn (c) matplotlib (d) numpy.

Q33

Why is NumPy generally faster than Python lists?

NumPy arrays are faster than Python lists because they use a fixed data type stored in contiguous memory locations, allowing computations to bypass the slow Python interpreter and run via optimized C code.

✓ Model Answer

NumPy arrays are contiguous fixed-type C arrays, so operations run in compiled C with SIMD vectorization. Python lists store pointers to heterogeneous Python objects — each op has interpreter overhead and pointer chasing.

Q34

Difference between list and numpy array.

Type your answer here...

✓ **Model Answer**

List: heterogeneous types, dynamic, slow elementwise math, no broadcasting. NumPy array: homogeneous dtype, fixed size, supports vectorized math and broadcasting, much less memory, much faster for numeric work.

Q35

What is broadcasting in NumPy? Give an example.

Broadcasting in NumPy is a mechanism that allows arithmetic operations to be performed on arrays of different shapes

✓ **Model Answer**

Broadcasting lets NumPy operate on arrays of different shapes by virtually stretching the smaller one. Example: a (3,3) matrix + a (3,) row vector adds the row to every row of the matrix without writing a loop. e.g. `np.array([[1,2,3],[4,5,6]]) + np.array([10,20,30])` → `[[11,22,33],[14,25,36]]`.

Section H: Jupyter & Development

Q36

Advantages of Jupyter Notebook over a normal Python script.

it works better in ml and algorithm

✓ **Model Answer**

Interactive cell-by-cell execution, inline plots and tables, markdown narration alongside code, great for EDA and demos, easy to share results with non-coders.

Q37

When should you avoid notebooks and move to Python modules?

Type your answer here...

✓ Model Answer

When building production pipelines, reusable code, libraries, code that needs unit tests, version control diffs, scheduled jobs, or anything multi-developer. Notebooks have hidden state, poor diffability, and don't scale to engineering workflows.

Q38

Difference between .ipynb and .py.

Type your answer here...

✓ Model Answer

.ipynb is a JSON file containing cells, outputs, and metadata — opened in Jupyter. .py is plain Python source code, runnable directly with ``python file.py``, importable as a module, friendly to git.

Q39

Why do companies keep notebooks in a separate folder?

Type your answer here...

✓ Model Answer

Notebooks are for exploration/EDA, not production. Separating them (notebooks/ vs src/) keeps the deployable codebase clean, makes imports clear, and signals which code is experimental vs production.

Q40

Purpose of src/, tests/, notebooks/, models/.

Type your answer here...

✓ **Model Answer**

src/ → production source code (data loaders, training, inference modules). tests/ → unit & integration tests. notebooks/ → exploratory analysis & experiments. models/ → saved trained model artifacts (often gitignored or stored in object storage / MLflow registry).

Section I: Production ML

Q41

Why do companies use Docker for ML projects?

Type your answer here...

✓ **Model Answer**

Packages code + dependencies + OS libs into a portable image — 'works on my machine' goes away. Same container runs in dev, staging, prod, and Kubernetes. Critical when models need specific Python / CUDA / library versions.

Q42

What problem does FastAPI solve?

Type your answer here...

✓ **Model Answer**

Exposes the ML model as a REST API quickly — clients send JSON, get predictions.

Async, fast, auto-generates OpenAPI docs, validates inputs with Pydantic. Standard way to serve scikit-learn / PyTorch models in production.

Q43

What is MLflow used for?

Type your answer here...

✓ **Model Answer**

Tracks ML experiments (parameters, metrics, artifacts), versions and registers models, packages models for deployment. Lets teams compare runs, reproduce results, and promote a model from staging to production.

Q44

Why is storing models on a laptop not production-ready?

Type your answer here...

✓ **Model Answer**

Not accessible to services, not versioned, no backup, no access control, single point of failure, no audit trail, can't serve concurrent requests. Production models live in model registries / object storage (MLflow, S3) and are served via APIs.

Q45

What is model deployment?

Type your answer here...

✓ **Model Answer**

The process of taking a trained model and making it available to end users / systems for predictions — wrapping it in an API, containerizing, hosting on a server / cloud / edge device, monitoring it, and updating it.

Section J: Practical Thinking

Q46

Churn model has 95% accuracy. What would you ask before deploying?

Type your answer here...

✓ Model Answer

What is class balance (95% could be the no-churn rate)? Precision/Recall/F1?
Performance on recent data? Bias across customer segments? Latency requirements?
How are predictions consumed (batch vs realtime)? Cost of FP vs FN? Drift monitoring plan? A/B test design?

Q47

Model good in training, poor in production — reasons?

Type your answer here...

✓ Model Answer

Overfitting; data leakage during training; training/serving skew (different preprocessing); data drift (production data $\neq$ training data); concept drift (relationship changed); feature pipeline bugs; different feature engineering at inference; class imbalance handled in training but not in metrics.

Q48

Predictions must be under 100 ms — considerations?

Type your answer here...

✓ **Model Answer**

Use lightweight models (logistic regression, small trees, distilled NN); reduce feature count; precompute / cache features (feature store); batch GPU inference; quantize / prune model; use ONNX or TensorRT; co-locate API and feature store; avoid heavy preprocessing at request time.

Q49

Model performance drops after 6 months — what happened?

Type your answer here...

✓ **Model Answer**

Data drift (input distribution changed), concept drift (target relationship changed), seasonality not modeled, broken upstream data pipeline, new product/customer segments, external events (covid-like shock), label quality drop. Need monitoring + scheduled retraining.

Q50

More important: (a) complex model or (b) solving the business problem?

Type your answer here...

✓ **Model Answer**

(b) Solving the business problem. A simple logistic regression that lifts revenue 5% beats a fancy deep net that's hard to deploy, explain, and maintain. Industry rewards business impact, not algorithmic sophistication. Start simple, add complexity only if it pays off.

Q51

Company says: 'Our revenue is decreasing. Can you build an ML model?' What do you ask first?

Type your answer here...

✓ Model Answer

Before any algorithm: (1) Business objective — what specifically should we change? (2) What is the current process / baseline? (3) Available data — what, where, quality, history? (4) What is the target variable — is the problem even an ML problem? (5) Success metric — business KPI and model metric. (6) Deployment constraints — latency, scale, infrastructure, batch vs realtime. (7) Expected business impact and ROI. (8) Stakeholders, timelines, regulatory/ethical constraints. The point: ML may not even be the answer — could be pricing, sales ops, marketing, or product. Problem framing > algorithm choice.

Submitted! Auto-graded MCQs: **2 / 2**. Open-ended answers — compare yours with the model answers shown in green below each question.

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